

Guidance Document: Validation of Rapid Diagnostics for Pathogen Identification and Antimicrobial Susceptibility Testing (AST) 2025

Preface

- This document has been designed to provide assistance to innovators and testing laboratories in validating diagnostics meant for pathogen identification and antimicrobial susceptibility testing.
- This validation protocol lays out a comprehensive framework to systematically evaluate and confirm the diagnostic performance of a given test, ensuring its reliability and its utility for clinical decision-making.
- This document describes evaluation criteria for methods to detect, identify, and quantify pathogenic micro-organisms or their genetic materials (DNA, RNA, toxins, antigens) as well as methods for antimicrobial susceptibility.
- It encompasses the assessment of precision, accuracy, reproducibility, and the ability to correctly identify the target pathogen and/or antimicrobial susceptibility.

Foreword

- Antimicrobial resistance (AMR) poses a significant global threat to efforts in combating infectious diseases.
- The inability to provide accurate and timely diagnosis is one of the major contributors to the misuse and overuse of antimicrobials.
- To address this gap, the Indian Council of Medical Research (ICMR), in consultation with the Central Drugs Standard Control Organisation (CDSCO), has developed a comprehensive protocol to guide the validation process for rapid diagnostic tests.

Chapter 1: Introduction

- Accurate diagnosis of the infection using a quality-assured diagnostic test is essential for the timely initiation of treatment and for reducing the misuse of antimicrobials.
- Validation of a diagnostic test is a systematic evaluation of the test which has been developed, standardized, and optimized, to determine its fitness for the specific intended use.
- ISO 15189:2022 and ISO 20916:2019 emphasize establishing the performance specifications as well as quality assessment measures as part of the validation.

Regulatory Process Followed by CDSCO

- For a new In-vitro Diagnostic (IVD) Medical Device without a predicate device, the applicant must obtain a Test License in form MD-13 to produce test batches.
- To generate clinical performance data, the applicant must secure permission to conduct Clinical Performance Evaluation (CPE) by obtaining MD-25.
- Study results and generated data must be submitted for approval to manufacture using form MD-29.
- Following this, the applicant must obtain the appropriate manufacturing license (MD-5/MD-6 or MD-9/MD-10) for sale or distribution.
- If a predicate device already exists in the country, the applicant obtains MD-13 for test batches, bypassing the CPE stage, and directly applies for the manufacturing license.

Chapter 2: General Considerations for Validation of Diagnostics

- **Sites:** Validation studies must preferably be carried out at multiple sites. If not feasible, validation should be carried out in a minimum of two different sites representing the country's geographic variations.
- **Sample:** The type of specimen to be used for testing must be mentioned (e.g., invasive or non-invasive). Isolates from sterile sites are more clinically significant than those from non-sterile sites. Sample sizes must be determined with the help of a statistical expert.

- **Test methods:** Specify the type of diagnostic, whether it is a method, assay, kit, or a device. Indicate the method involved (e.g., biochemical, molecular, or mass-spectrometry).
- **Standards:** Reference material for validating the test must be of certified quality. Strains used to measure test performance should be obtained from reference centre collections (e.g., ATCC).
- **Standards for Interpretation:** Breakpoints are set by the Clinical and Laboratory Standards Institute (CLSI) and the European Committee on Antimicrobial Susceptibility Testing (EUCAST). Minimum Inhibitory Concentration (MIC) is the lowest concentration of an antibiotic which prevents visible growth of a bacterium.
- **Reference method:** The reference method is usually the gold standard for the organism under study. Validation studies must include comparison to a recognized reference method to demonstrate equivalence of performance.
- **Media and Supplies:** ISO standard bacteriological media should be utilized for antimicrobial susceptibility testing. Reagents and kits must be supplied at a temperature that is suitable to preserve test efficacy, and storage conditions must be clearly stated.

Chapter 3: Assay Development and Validation Criteria

Phase 1: Assay Development

- **Step 1: Define the Purpose of Validation:** Outline goals such as pathogen identification or antimicrobial susceptibility testing. Determine whether the test provides a qualitative or quantitative result, and if a screening or confirmatory test is required.
- **Step 2: Optimization of Reagents and Protocol:** Reagents and protocols must be optimized to ensure consistency across temporal, chemical, and physical variables. A kit should be self-sufficient to allow testing and analysis.
- **Step 3: Quality Control (QC) and Assurance (QA):** The test and reference method should pass defined QC criteria before validation. Up to nine replicates should be tested for the reference method, including at least one QC organism. Microbial reference strains should be obtained from standard sources.

Phase 2: Assay Validation

- **Step 4: Analytical Performance SOPs:** Document the study design, sample size, and settings. For pathogen identification, a representative number of positive and negative samples should be tested in parallel. For AST, a minimum of 100 isolates recovered from clinical samples are recommended for each group of micro-organisms.
- **Step 5: Testing Performance Parameters:**
 - **Pathogen Identification:** Evaluate Accuracy, Analytical Sensitivity (distinguishing close concentrations), Analytical Specificity (interfering substances), Reproducibility, Precision, Reportable range, and Reference range.
 - **Antimicrobial Susceptibility Testing (AST):** Assess Accuracy via Categorical Agreement ($\geq 90\%$) and Essential Agreement ($\geq 90\%$). Evaluate Error Rates including Minor Errors ($< 10\%$), Major Errors ($< 3\%$), and Very Major Errors ($< 3\%$). Assess Precision and Reproducibility across multiple runs and days.
- **Step 6: Clinical Validation:** Clinical performance evaluation is the systematic evaluation of a new in vitro diagnostic device using specimens collected from human participants. Evaluate Clinical Validity (ability to detect the disorder) and Clinical Utility (usefulness in diagnosis or treatment). Calculate Clinical Sensitivity and Clinical Specificity. Ethical approval and informed consent must be obtained.

Phase 3: Assay Maintenance

- **Step 7: Maintenance and Extension:** A validated assay needs constant monitoring and maintenance. Internal quality control monitors repeatability and accuracy, while external control utilizes proficiency testing.
- **Step 8: Dossier of Validation:** Documentation of all validation and verification experiments must be kept by the laboratory for no less than 2 years. It should contain testing records, method modifications, and justifications for inadequate data.
- **Step 9: Outcome and Decision:** The decision on the outcome of the validation process depends on the practicability of the method, assay usability, performance parameters data, and user feedback.